

Dr. Zhiwei Huang

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SKILLS

- **MR specialty:** pulse sequence development, image reconstruction, MRI physics, MR acquisitions, spectral processing, volunteer coordination.
- **Data Processing Pipeline Development:** End-to-end MRS/MRSI acquisition-to-analysis workflows; DICOM data handling; quality control (QC); reproducibility validation; deployment of fMRI/fMRSI processing pipelines; machine learning (regression, classification); deep learning (variational neural network, CNN)
- **Scientific Communication:** Study reporting, manuscript preparation, interdisciplinary collaboration, presentation of findings to scientific stakeholders.
- **Programming:** MATLAB, Python, Siemens IDEA (C++), Git.
- **Languages:** Mandarin (native), English (professional proficiency; TOEFL 108), French (basic, A1)

SELECTED PROFESSIONAL EXPERIENCES

Functional brain metabolic mapping with magnetic resonance spectroscopic imaging (MRSI)

- **Sequence optimization, clinical acquisitions:** Developed 2D/3D RSI sequence for functional applications; designed and executed fMRI/fMRSI experiments on healthy volunteers.
- **Image analysis, statistical analysis:** Developed a full analysis pipeline for functional metabolic images, including image reconstruction, spectral quantification, quality control, and statistical modelling (GLM/ICA), as well as fMRI analysis for comparison (data size per subject > 80G/subject).
- **Interdisciplinary collaboration:** Collaborated with spectroscopists and neuroscientists to interpret metabolic activation maps and align imaging findings with functional hypotheses.
- **Output:** Huang Z. et al., Mapping Glutamate Responses to Visual Stimulation with Functional Magnetic Resonance Spectroscopic Imaging. *ISMRM 2026*, South Africa, Oral.

Fast MRSI sequence and reconstruction pipeline development with rosette trajectories (CIBM)

- **Sequence development:** Developed 2D/3D fast rosette MRSI sequences from scratch in Siemens IDEA (VB17, XA60) and validated them *in vitro* and *in vivo*. Improved water suppression efficiency by 5 times through Bloch simulation based water suppression block optimization.
- **End-to-end signal/image analysis pipeline development:** Adapted non-Cartesian 2D-/3D-*t* image reconstruction algorithms, integrating NUFFT and compressed sensing to achieve 3× acceleration while preserving spectral and spatial fidelity. Performed anatomical image segmentation and parcellation for T_1 relaxation and water content correction, as well as region of interest analysis. Optimized an L_2 lipid removal algorithm with 50 times larger lipid contaminations.
- **Output:** Huang Z, et al. Rosette Spectroscopic Imaging for Whole-Brain Slab Metabolite Mapping at 7T: Acceleration Potential and Reproducibility. *Hum Brain Mapp.* 2025;46(4):e70176.

³¹P whole-brain MRSI sequence optimization and post-processing (CIBM)

- **MR sequence optimization:** Optimized a conventional chemical shift imaging (CSI) sequence (VB17) for ³¹P acquisition and successfully acquired whole-brain MRSI data targeting NAD⁺/NADH in the human brain.
- **Image reconstruction:** Constructed a full ³¹P Cartesian MRSI processing pipeline, including reconstruction, spectral fitting, and metabolic map visualization, enabling spatially resolved quantification of NAD⁺/NADH.
- **Output:** Huang, Z. et al., Whole brain mapping of NAD at 7T using a 31P 32 channel array coil. *ISMRM 2024*, Singapore, power pitch.

³¹P diffusion MR spectroscopy method development (CIBM)

- **Sequence development:** Developed a custom ³¹P diffusion MRS sequence in Siemens IDEA (VB17).
- **Data analysis:** Developed a full analysis pipeline including coil combination, quality control, and ADC fitting. Performed ANOVA analysis for PCr ADC anisotropy characterization.
- **Output:** Huang, Z., Gambaratto, G., Xiao, Y., Wenz, D., & Xin, L. (2023). Apparent diffusion coefficients of ³¹P metabolites in the human calf muscle at 7T.

Exploration of circadian rhythm's influence on brain metabolites (CIBM & Nestle)

- **Clinical acquisitions:** Conducted a clinical-style cohort study on 25 human participants with repeated MRI/MRS acquisitions over time. Responsibilities included protocol planning, volunteer coordination, image acquisition, data QC, statistical analysis, and scientific reporting.
- **Data analysis and statistical analysis:** Processed and quantified MRS spectra using model-based fitting; evaluated circadian effects on metabolite levels quantitatively through statistical analysis.
- **Output:** Cuenoud, B., Huang, Z. et al. (2023). Effect of circadian rhythm on NAD and other metabolites in human brain. *Frontiers in Physiology*, 14, 1285776.

Fast MRI reconstruction using variational neural network (ETHz)

Master thesis

- **Algorithm design:** Designed an image sharpness metric for quantitative image quality assessment; and a spectral-domain loss function ($exp-L_1$) tailored to enhance edge preservation and reconstruction sharpness, yielding higher SSIM compared to standard L_1/L_2 losses.
- **Neural network optimization and benchmarking:** Optimized the hyperparameters for a VNN and benchmarked multiple loss functions (TensorFlow), achieving SSIM > 0.8 at 4× and > 0.7 at 8× acceleration.

INTERNSHIPS

Device development intern, Novartis International AG (Basel, Switzerland)

05/2020 – 10/2020

- **Software development:** Built a *Python*-based automation tool to streamline reports *Polarion* (a software), and created a *JavaScript* widget to extract metadata from *Velocity* (a software).
- **Database development and visualization:** Constructed a database and developed a *Python*-based web dashboard for interactive data querying, visualization, and basic statistical analysis.
- **Feedback:** Supervisor evaluation: excellent, with independence, quick learning, and effective team integration.

EDUCATION

École polytechnique fédérale de Lausanne (EPFL)

Lausanne, Switzerland

Doctor student in EDEE (Electrical Engineering), CIBM (Center for Biomedical Imaging)

07/2021 – present

- Thesis: Functional magnetic resonance spectroscopic imaging method development at 7T.

École polytechnique fédérale de Lausanne (EPFL)

Lausanne, Switzerland

Master degree (Major: Life Science Engineering: Biomedical Image Analysis)

09/2018 – 06/2021

- GPA: 5.53/6.00
- Thesis: MRI reconstruction with variational neural networks (CMR lab, ETHz).

University of Chinese Academy of Sciences (UCAS)

Beijing, China

Bachelor degree (Major: Materials Science and Engineering; Minor: Biology)

09/2014 – 08/2018

- GPA: 3.73/4.00, Ranking: 5/49 (major); 37/114.00 (minor)
- Excellent Graduates in Beijing (Top 2%, 2018)

Columbia University in the City of New York

New York, USA

Exchange student (Materials Science and Engineering)

01/2017 – 05/2017

- GPA: 4.15/4.33

TEACHING

Student project supervision: Brain metabolic mapping by functional MRS/I	Spring 2025
Student project co-supervision: ^{31}P MRS denoising using CNN	Autumn 2023
Teaching assistant: Neural signal and signal processing	Fall, 2023-2025,
Teaching assistant: Fundamentals of Biomedical Imaging	Spring, 2023
Teaching assistant: Image Processing I	Spring, 2022

PUBLICATIONS

- [1] **Huang, Z.**, Emir, U., Doring, A., Klauser, A., Xiao, Y., Widmaier, M., & Xin, L. (2024). Rosette spectroscopic imaging for whole-brain metabolic mapping at 7T: acceleration potential and reproducibility. *Human Brain Mapping*, 2025 Mar;46(4):e70176. doi: 10.1002/hbm.70176. PMID: 40056040; PMCID: PMC11889463.
- [2] **Huang, Z.**, Gambaratto, G., Xiao, Y., Wenz, D., & Xin, L. (2023). Apparent diffusion coefficients of ^{31}P metabolites in the human calf muscle at 7T. *Magnetic Resonance Materials in Physics, Biology and Medicine*, 36(2), 395-411.
- [3] Cuenoud, B., **Huang, Z.**, Hartweg, M., Widmaier, M., Lim, S., Wenz, D., & Xin, L. (2023). Effect of circadian rhythm on NAD and other metabolites in human brain. *Frontiers in Physiology*, 14, 1285776.
- [4] Widmaier, M. S., Kaiser, A., Baug, S., Wenz, D., Pierzchala, K., Xiao, Y., **Huang, Z.**, & Xin, L. (2025). Fast 3D ^{31}P B1+ mapping with a weighted stack of spiral trajectory at 7T. *Magnetic Resonance in Medicine*, 93(2), 481-489.
- [5] Xiao, Y., Kaiser, A., Kockisch, M., Back, A., Carlet, R., Liu, X., ... & Xin, L. (2025). A graphical pipeline platform for MRS data processing and analysis: MRspecLAB. *Front. Neuroimaging* 4:1610658. doi: 10.3389/fnimg.2025.1610658
- [6] Widmaier, M., Kaiser, A., Pandurevic, P., Lim, S. I., Döring, A., **Huang, Z.**, ... & Xin, L. (2025). Creatine Kinase Imaging (CKI) for In Vivo Whole-Brain Mapping of Creatine Kinase Reaction Kinetics. *Proceedings of the National Academy of Sciences*. In press.

CONFERENCES

- [1] **Huang, Z.**, Doring A., Antonia, K., Penghui D., ..., & Xin, L. Mapping Glutamate Responses to Visual Stimulation with Functional Magnetic Resonance Spectroscopic Imaging. *ISMRM 2026, South Africa*, Oral.
- [2] **Huang, Z.**, Emir, U., Doring, A., Klauser, A., Xiao, Y., Widmaier, M., & Xin, L. Rosette spectroscopic imaging for whole-brain metabolic mapping at 7T: acceleration potential & reproducibility. *MRS workshop 2022*, Boston, poster.
- [3] **Huang, Z.**, Cuenoud, B., Hartweg, M., Wenz, D., Xin, L. Effect of circadian rhythm on brain NAD: an MRS study at 7T. *ISMRM-ESMRMB 2022*, U.K., poster.
- [4] **Huang, Z.**, Cuenoud, B., Hartweg, M., Wenz, D., Xin, L. Effect of circadian rhythm on brain NAD: an MRS study at 7T. *MRS workshop 2022*, Switzerland, Oral presentation.
- [5] **Huang, Z.**, Widmaier, M., Wenz, D., Emir, U., Xin, L. Whole brain mapping of NAD at 7T using a ^{31}P 32 channel array coil. *ISMRM 2024*, Singapore, power pitch.
- [6] **Huang, Z.**, Gambaratto, G., Xiao, Y., Wenz, D., Xin, L. Apparent diffusion coefficients of ^{31}P metabolites in the human calf muscle at 7T. *ISMRM 2022*, Canada, poster. *MRS workshop 2024*, U.S., poster.
- [7] Doring, A., **Huang, Z.**, Kaiser, A., Landheer, K., Emir, U., & Xin, L. *Functional Glutamate and GABA response maps to visual stimulation by functional magnetic resonance spectroscopic imaging*. ISMRM 2025, Hawaii, Oral presentation.
- [8] Doring, A., **Huang, Z.**, Kaiser, A., Landheer, K., Emir, U., Xin, L. Optimization of 2D rosette spectroscopic imaging for mapping brain neurotransmitters at 7T. *ESMRMB 2024*, Spain, poster.